Praktikum Elektronika Analog

Feedback Voltage-Series dan Current-Series

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Tujuan Praktikum

1. Mengetahui parameter yang dibutuhkan pada datasheet untuk rangkaian feedback
2. Mampu merangkai rangkaian feedback voltage-series dan current-series pada konfigurasi common emitter BJT di protoboard
3. Mampu melakukan perhitungan penguatan feedback pada rangkaian feedback
4. Mampu melakukan pengukuran parameter yang dibutuhkan untuk pembuktian rangkaian feedback

Peralatan yang dibutuhkan

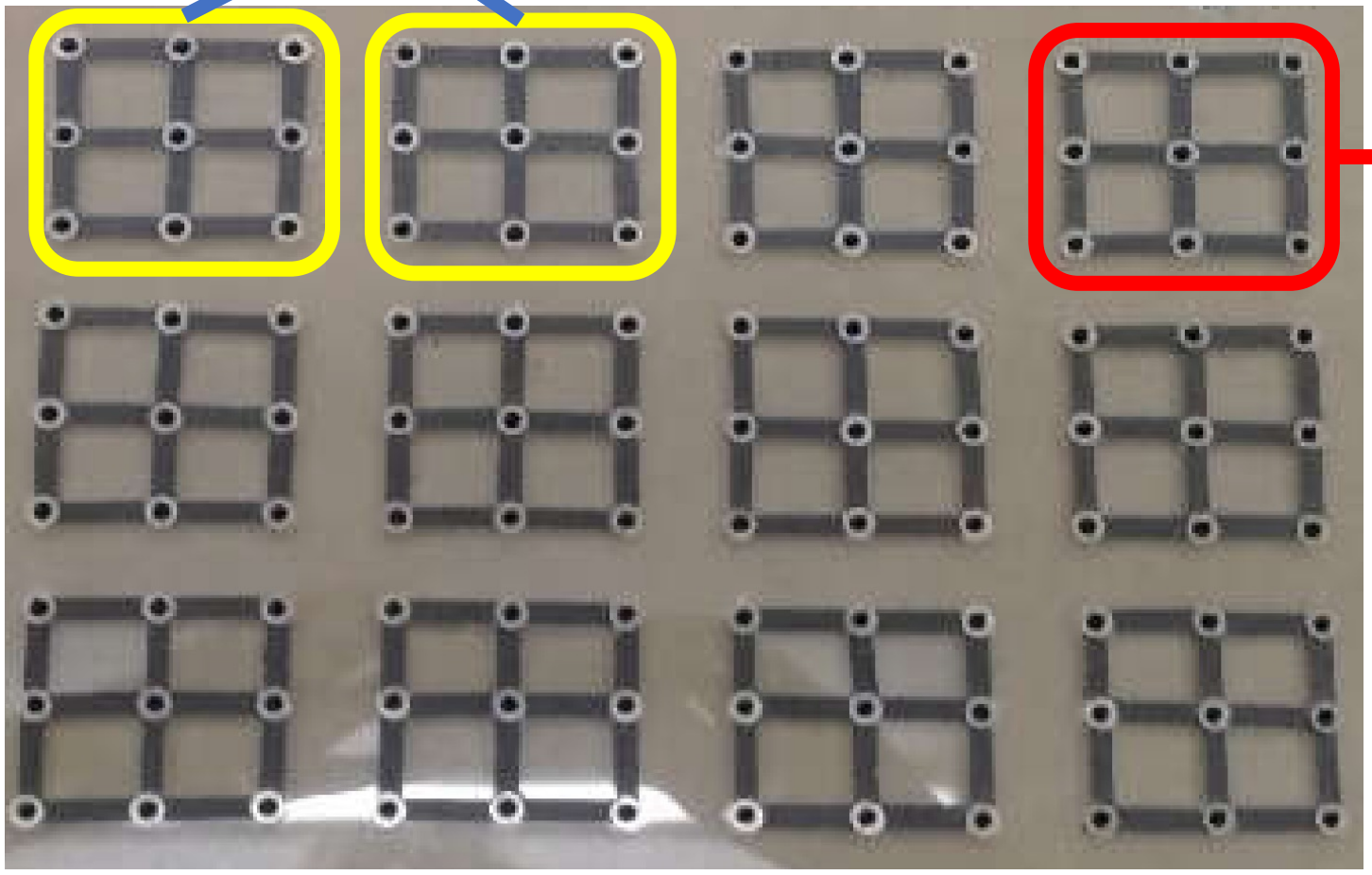
1. DC Power Supply (1)
2. Function Generator (1)
3. Oscilloscope (1)
4. BC107/BC108 NPN Transistor atau padanannya TO-92 (1)
5. R_B= 470KOhm; R_C= 2KOhm; R_E= 1.2KOhm
6. $C_1 = C_2$ = 10µF (2)
7. Multimeter (2)
8. Kabel Jumper (secukupnya)
9. Konektor Jumper (secukupnya)
10. Kertas HVS dan millimeter block (masing-masing 1 lembar/kelompok)

Persiapan praktikum yang harus dilakukan

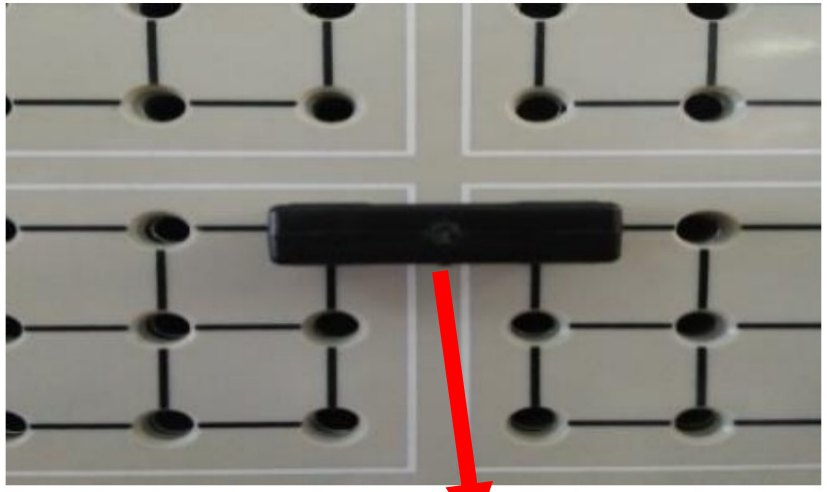
1. Melakukan pengecekan kabel
2. Melakukan pengecekan transistor
3. Merangkai rangkaian percobaan dengan benar sesuai dengan desain yang telah dibuat
4. Memperhatikan dan menerapkan K3

Pengenalan Protoboard

Tidak saling terhubung

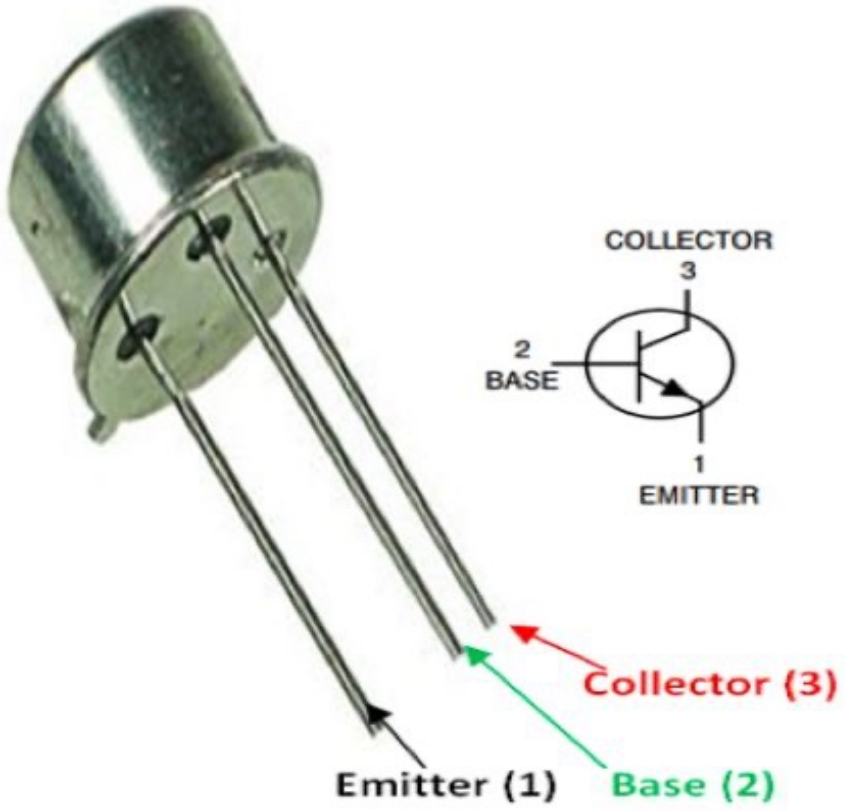


Semua titik saling terhubung



Contoh menghubungkan
2 kelompok matrix dengan jumper/konektor

Cara mengecek BJT dengan multimeter



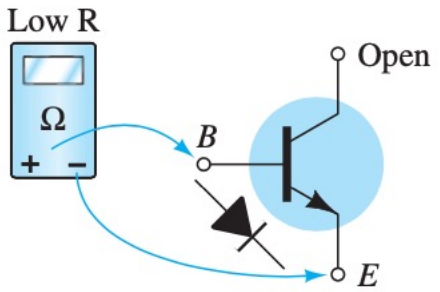


FIG. 3.27
Checking the forward-biased base-to-emitter junction of an npn transistor.

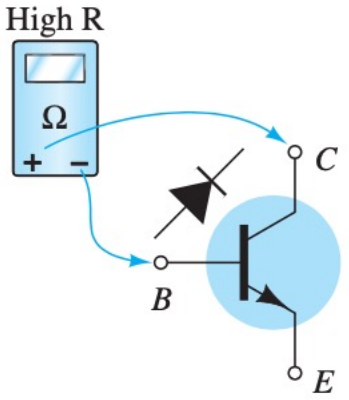


FIG. 3.28
Checking the reverse-biased base-to-collector junction of an npn transistor.

Datasheets (1/3)

BC107/BC108 Series

Low Power Bipolar Transistors

multicomp

Absolute Maximum Ratings

Description	Symbol	BC107	BC108	Unit
Collector-Emitter Voltage	V_{CEO}	45	25	V
Collector-Base Voltage	V_{CBO}	50	30	
Emitter-Base Voltage	V_{EBO}	6.0	5.0	
Collector Current Continuous	I_C	0.2		A
Power Dissipation at T_a = 25°C Derate above 25°C	P_D	0.6 2.28		W mW/°C
Power Dissipation at T_C = 25°C Derate above 25°C		1.0 6.67		
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-65 to +200		°C
Thermal Resistance				
Junction to Case	$R_{th\ (j-c)}$	175		°C/W

Datasheets (2/3)

Electrical Characteristics (T_a = 25°C unless otherwise specified)

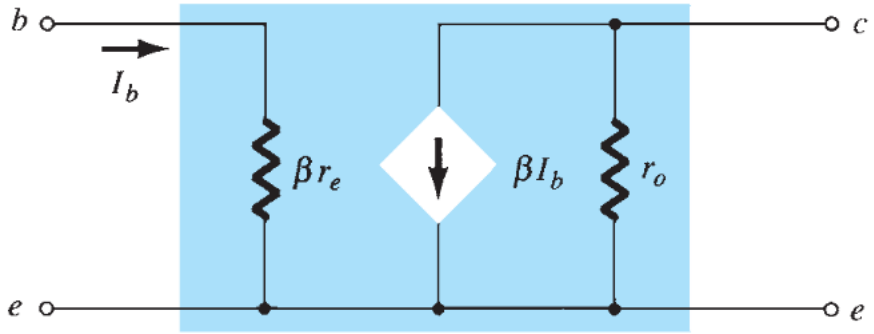
Description	Symbol	Test Condition	Minimum	Maximum	Unit
Collector-Emitter Voltage	V_{CEO}	I_C = 2mA, I_B = 0 BC107 BC108	45 25	-	V
Emitter-Base Voltage	V_{EBO}	I_E = 10µA, I_C = 0 BC107 BC108	6.0 5.0	-	V
Collector-Cut off Current	I_{CBO}	V_{CB} = 45V, I_E = 0 BC107 V_{CB} = 25V, I_E = 0 BC108 T_{amb} = 125°C V_{CB} = 45V, I_E = 0 BC107 V_{CB} = 25V, I_E = 0 BC108	-	15 15 4.0 4.0	nA µA
DC Current	h_{FE}	I_C = 10µA, V_{CE} = 5V **B Group** **C Group** I_C = 2mA, V_{CE} = 5V **BC 107** **BC 108** **A Group** **B Group** **C Group**	40 100 110 110 110 200 420	- - 450 800 220 450 800	-
Base Emitter Saturation Voltage	$V_{BE\ (Sat)}$	I_C = 10mA, I_B = 0.5mA I_C = 100mA, I_B = 5mA	-	0.83 1.05	V
Collector Emitter Saturation Voltage	$V_{CE\ (Sat)}$		-	0.25 0.60	V
Base Emitter On Voltage	$V_{BE\ (on)}$	I_C = 2mA, V_{CE} = 5V I_C = 10mA, V_{CE} = 5V	0.55 -	0.70 0.77	V

Datasheets (3/3)

Electrical Characteristics (T_a = 25°C unless otherwise specified)

Description	Symbol	Test Condition	Minimum	Maximum	Unit
Collector Knee Voltage	$V_{CE\ (K)}$	I_C = 10mA, I_B = The value for which I_C = 11mA at V_{CE} = 1V	-	0.60	V
Transition Frequency	f_t	V_{CE} = 5V, I_C = 10mA, f = 100MHz	150	-	MHz
Noise Figure	NF	V_{CE} = 5V, I_C = 0.2mA R_g = 2kΩ F = 1kHz,B = 200Hz		10	dB
Output Capacitance	C_{obo}	V_{CB} = 10V, f = 1MHz	-	4.5	pF
Small Signal Current Gain	h_{fe}	ALL f = 1kHz I_C = 2mA, V_{CE} = 5V **BC 107** **BC 108** **A Group** **B Group** **C Group**	 125 125 125 240 450	 500 900 260 500 900	-
Input Impedance	h_{ie}	I_C = 2mA, V_{CE} = 5V **A Group** **B Group** **C Group**	 1.6 3.2 6.0	 4.5 8.5 15	 KΩ KΩ
Output Admittance	h_{oe}	I_C = 2mA, V_{CE} = 5V **A Group** **B Group** **C Group**	-	 30 60 110	umhos

Rangkaian Ekivalen BJT Common Emitter



$$r_e = 26\,\text{mV}/I_E$$

$$Z_i = \frac{V_i}{I_b} = \frac{V_{be}}{I_b}$$

$$V_{be} = I_e r_e = (I_c + I_b) r_e = (\beta I_b + I_b) r_e$$
$$= (\beta + 1) I_b r_e$$

$$Z_i = \frac{V_{be}}{I_b} = \frac{(\beta + 1) I_b r_e}{I_b}$$

$$\boxed{Z_i = (\beta + 1) r_e \cong \beta r_e}$$

Resistor Color Code

Quick Guide

1st Band (F)
2nd Band (S)
Multiplier (M)

Color	Value
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Gray	8
White	9

Band 3	Divide by
Gold	10
Silver	100

4- Band Code



100 Ω ± 5%

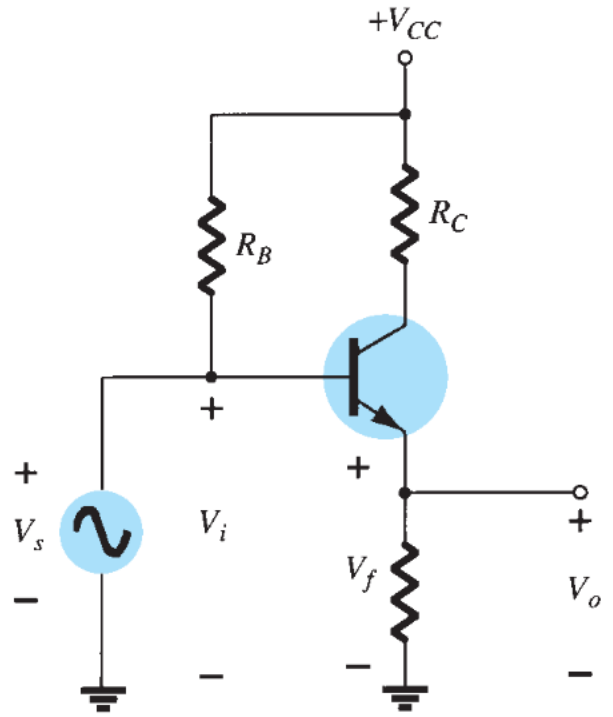
OR

Tolerance	
Brown	± 1 %
Red	± 2 %
Gold	± 5 %
Silver	± 10 %
None	± 20 %

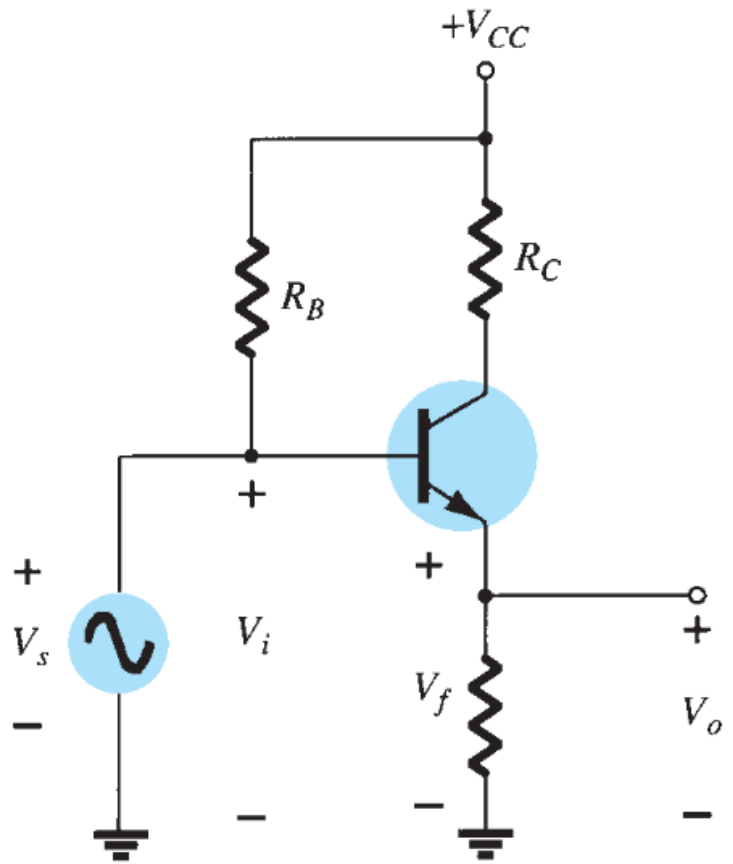
F S M								
.22 ohm								
.27 ohm								
.33 ohm								
.39 ohm								
.47 ohm								
.56 ohm								
.68 ohm								
.82 ohm								
1.0 ohm	10 ohm	100 ohm	1.0 k	10 k	100 k	1 M	10 M	
1.1 ohm	11 ohm	110 ohm	1.1 k	11 k	110 k	1.1 M		
1.2 ohm	12 ohm	120 ohm	1.2 k	12 k	120 k	1.2 M		
1.3 ohm	13 ohm	130 ohm	1.3 k	13 k	130 k	1.3 M		
1.5 ohm	15 ohm	150 ohm	1.5 k	15 k	150 k	1.5 M		
1.6 ohm	16 ohm	160 ohm	1.6 k	16 k	160 k	1.6 M		
1.8 ohm	18 ohm	180 ohm	1.8 k	18 k	180 k	1.8 M		
2.0 ohm	20 ohm	200 ohm	2.0 k	20 k	200 k	2.0 M		
2.2 ohm	22 ohm	220 ohm	2.2 k	22 k	220 k	2.2 M		
2.4 ohm	24 ohm	240 ohm	2.4 k	24 k	240 k	2.4 M		Resistor Values
2.7 ohm	27 ohm	270 ohm	2.7 k	27 k	270 k	2.7 M		
3.0 ohm	30 ohm	300 ohm	3.0 k	30 k	300 k	3.0 M		
3.3 ohm	33 ohm	330 ohm	3.3 k	33 k	330 k	3.3 M		
3.6 ohm	36 ohm	360 ohm	3.6 k	36 k	360 k	3.6 M		
3.9 ohm	39 ohm	390 ohm	3.9 k	39 k	390 k	3.9 M		
4.3 ohm	43 ohm	430 ohm	4.3 k	43 k	430 k	4.3 M		
4.7 ohm	47 ohm	470 ohm	4.7 k	47 k	470 k	4.7 M		
5.1 ohm	51 ohm	510 ohm	5.1 k	51 k	510 k	5.1 M		
5.6 ohm	56 ohm	560 ohm	5.6 k	56 k	560 k	5.6 M		
6.2 ohm	62 ohm	620 ohm	6.2 k	62 k	620 k	6.2 M		
6.8 ohm	68 ohm	680 ohm	6.8 k	68 k	680 k	6.8 M		
7.5 ohm	75 ohm	750 ohm	7.5 k	75 k	750 k	7.5 M		
8.2 ohm	82 ohm	820 ohm	8.2 k	82 k	820 k	8.2 M		
9.1 ohm	91 ohm	910 ohm	9.1 k	91 k	910 k	9.1 M		

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Rangkaian Skematik Percobaan 1 V-Series



Voltage-series feedback circuit



$$A = \frac{V_o}{V_s} = \frac{h_{fe} I_b R_E}{V_s} = \frac{h_{fe} R_E (V_s/h_{ie})}{V_s} = \frac{h_{fe} R_E}{h_{ie}}$$

$$\beta = \frac{V_f}{V_o} = 1$$

The operation with feedback then provides that

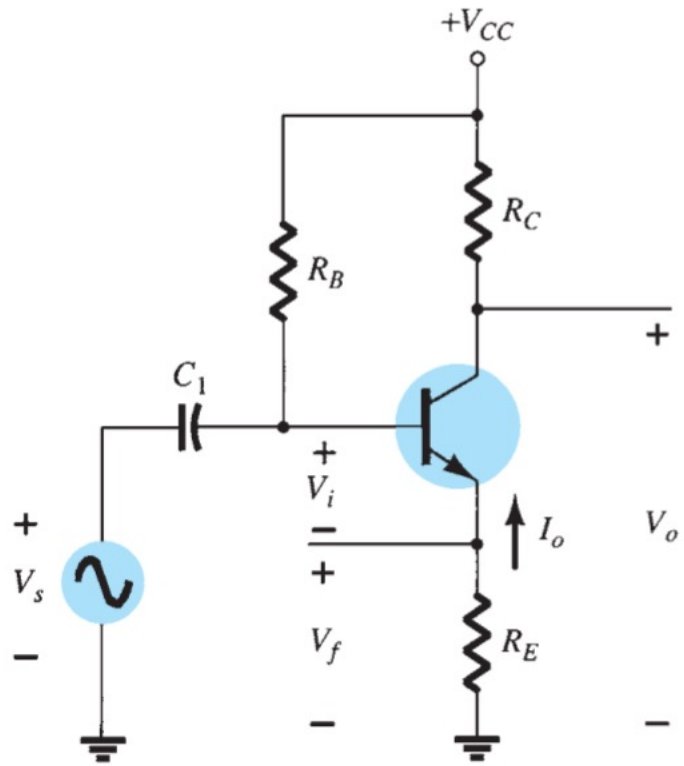
$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + \beta A} = \frac{h_{fe} R_E / h_{ie}}{1 + (1)(h_{fe} R_E / h_{ie})}$$

$$= \frac{h_{fe} R_E}{h_{ie} + h_{fe} R_E}$$

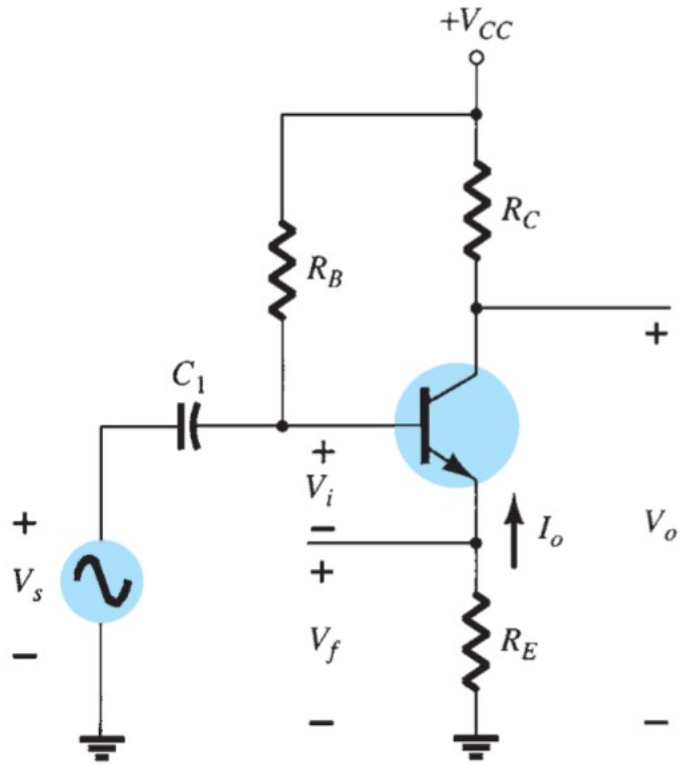
For $h_{fe} R_E \gg h_{ie}$,

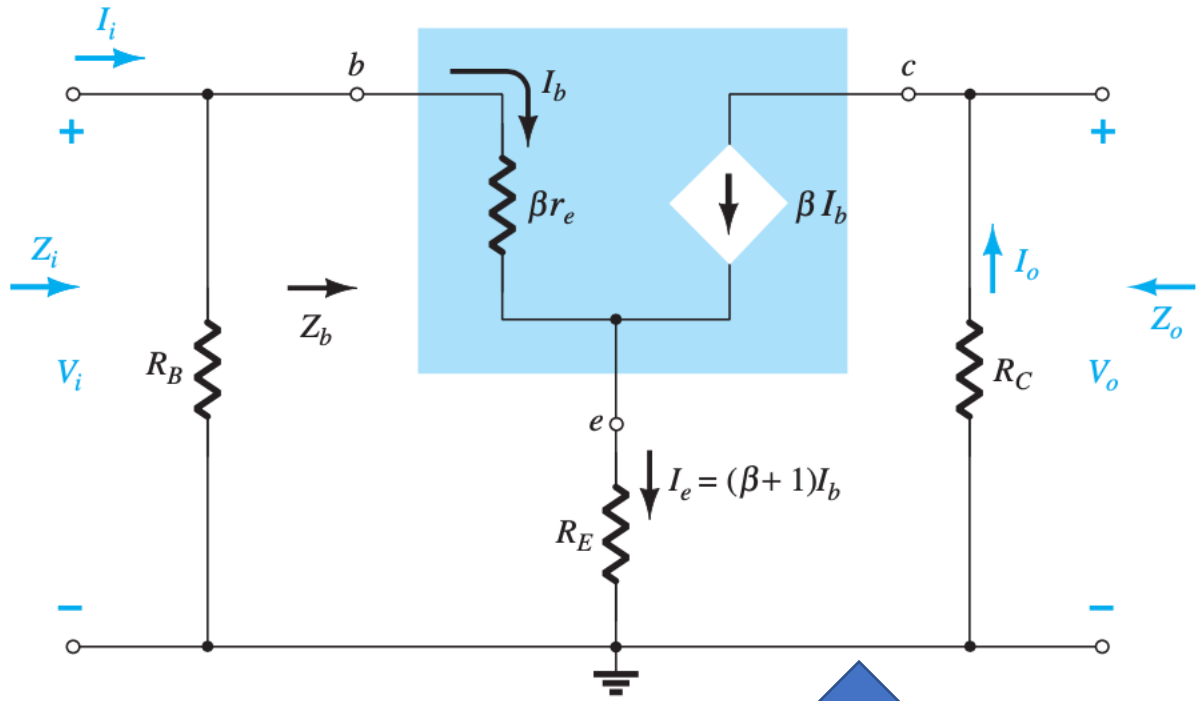
$$A_f \cong 1$$

Rangkaian Skematik Percobaan 2 C-Series



Current-series feedback





$$A = \frac{I_o}{V_i} = \frac{-I_b h_{fe}}{I_b h_{ie} + R_E} = \frac{-h_{fe}}{h_{ie} + R_E}$$

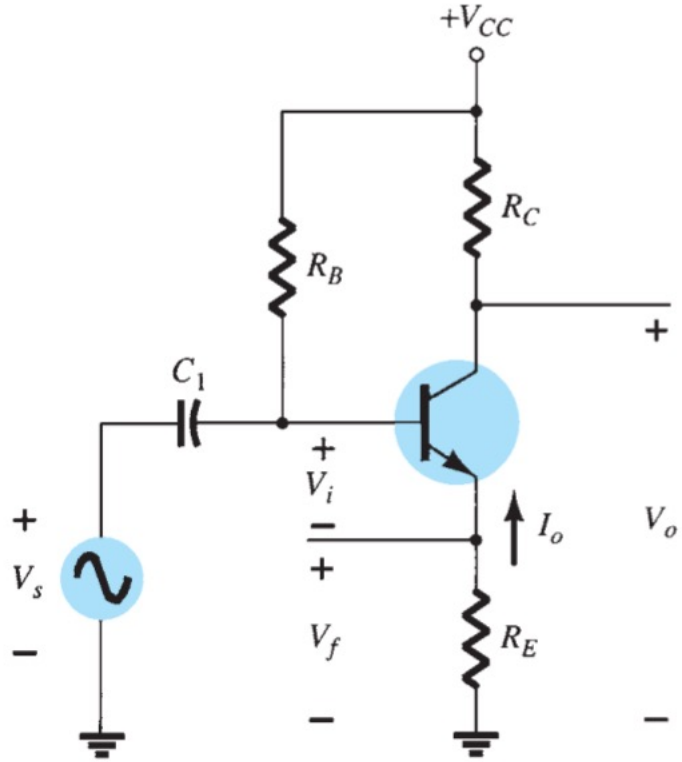
$$\beta = \frac{V_f}{I_o} = \frac{-I_o R_E}{I_o} = -R_E$$

The input and output impedances are, respectively,

$$Z_i = R_B \| (h_{ie} + R_E) \cong h_{ie} + R_E$$

$$Z_o = R_C$$

Current-series feedback



With Feedback

$$A_f = \frac{I_o}{V_s} = \frac{A}{1 + \beta A} = \frac{-h_{fe}/h_{ie}}{1 + (-R_E)\left(\dfrac{-h_{fe}}{h_{ie} + R_E}\right)} \cong \frac{-h_{fe}}{h_{ie} + h_{fe}R_E}$$

The input and output impedances are calculated as specified in Table 14.2:

$$Z_{if} = Z_i(1 + \beta A) \cong h_{ie}\left(1 + \frac{h_{fe}R_E}{h_{ie}}\right) = h_{ie} + h_{fe}R_E$$

$$Z_{of} = Z_o(1 + \beta A) = R_C\left(1 + \frac{h_{fe}R_E}{h_{ie}}\right)$$

The voltage gain A with feedback is

$$A_{vf} = \frac{V_o}{V_s} = \frac{I_o R_C}{V_s} = \left(\frac{I_o}{V_s}\right)R_C = A_f R_C \cong \frac{-h_{fe}R_C}{h_{ie} + h_{fe}R_E}$$

Langkah dan Tugas Percobaan

1. Rangkaialah komponen dan perangkat yang dibutuhkan sesuai dengan rangkaian percobaan 1
2. Pasang C1 dan C2 masing-masing di sisi input dan output untuk memblokir sinyal DC
3. Atur V_{CC} = 15V, Ukur nilai I_{CQ} dan V_{CEQ}
4. Berikan input AC sebesar 10mVpp dan f = 1KHz amati sinyal input dan output menggunakan oscilloscope
5. Gambarkan sinyal input dan sinyal output pada kertas milimeter block
6. Hitunglah penguatan yang terjadi berdasarkan pengukuran
7. Bandingkan pengutan tegangan berdasarkan perhitungan dan pengukuran; hitunglah %error yang terjadi
8. Rangkaialah komponen dan perangkat yang dibutuhkan sesuai dengan rangkaian percobaan 2
9. Ulangi dan lakukan percobaan seperti Langkah 2-7